



SANDIP
UNIVERSITY

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Neelam Vidya Vihar, Sijoul, Mailam, Madhubani, Bihar – 847235

Website: <http://www.sandipuniversity.edu.in> Email: info@sandipuniversity.edu.in

Program: B. Tech in Computer Science and Engineering (AI & ML)	
Year : Second Year	Semester : III
Course Title : Introduction to Machine Learning	Short Title : IML
Course Code : CSAIML304	
Theory : 3 Hours/Week	Tutorial: 0 Hour/Week
Continuous Internal Assessment: 50 Marks	Maximum University Theory Exam: 50 Marks
Time of Theory Exam: 2.5 hrs	Credit : 3

Course Objectives :

1	Understand fundamental machine learning concepts, learning paradigms, and dataset handling techniques including feature engineering and validation methods.
2	Implement supervised learning algorithms for classification and regression tasks, and evaluate model performance using appropriate metrics.
3	Apply unsupervised learning techniques including clustering algorithms to discover patterns in unlabeled datasets.
4	Utilize dimensionality reduction methods and advanced machine learning models to solve complex data analysis problems.
5	Explore cutting-edge machine learning domains including deep learning, NLP and computer vision applications.

Course Outcomes :

1	Explain fundamental machine learning concepts, dataset handling, and evaluation techniques (cross-validation, feature sets).
2	Implement supervised learning algorithms and evaluate models using metrics.
3	Apply unsupervised learning techniques to analyze unlabeled datasets.
4	Utilize dimensionality reduction methods and advanced models for data preprocessing and pattern recognition.
5	Explore emerging domains and deep learning basics to solve real-world problems.

Unit No.	Details	Hours
1	Introduction: What Is Machine Learning?, How Do We Define Learning?, How Do We Evaluate Our Networks?, How Do We Learn Our Network?, What are datasets and how to handle them?, Feature sets, Dataset division: test, train and validation sets, cross validation.	8
2	Basics of machine learning:	8

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	Applications of Machine Learning, processes involved in Machine Learning, Introduction to Machine Learning Techniques: Supervised Learning, Unsupervised Learning and Reinforcement Learning, Real life examples of Machine Learning.	
3	Supervised learning: Classification and Regression: K-Nearest Neighbor, Linear Regression, Logistic Regression, Support Vector Machine (SVM), Evaluation Measures: SSE, MME, R2, confusion matrix, precision, recall, F-Score, ROC-Curve.	8
4	Unsupervised learning: Introduction to clustering, Types of Clustering: Hierarchical, Agglomerative Clustering and Divisive clustering; Partitional Clustering - K-means clustering.	8
5	Miscellaneous: Dimensionality reduction techniques: PCA, LDA, ICA. Introduction to Deep Learning, Gaussian Mixture Models, Natural Language Processing, Computer Vision.	8
Total (Hrs)		40

Resources :

Text Books & Reference Books	1. Introduction to Machine Learning, By Jeeva Jose, Khanna Book Publishing Co., 2020. 2. Machine Learning for Dummies, By John Paul Mueller and Luca Massaron, For Dummies, 2016. 3. Machine Learning, By Rajeev Chopra, Khanna Book Publishing Co., 2021. 4. Machine Learning: The New AI, By Ethem Alpaydin, The MIT Press, 2016. 5. Machine Learning, Tom M. Mitchell, McGraw Hill Education, 2017.
E-Resources	1. https://www.udacity.com/course/intro-to-machine-learning--ud120 2. https://www.coursera.org/learn/machine-learning-duce 3. http://nptel.ac.in/

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